

Surveillance techniques evaluation, a computer science and public law study

From an example

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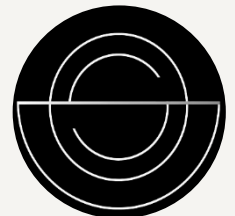
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Summary

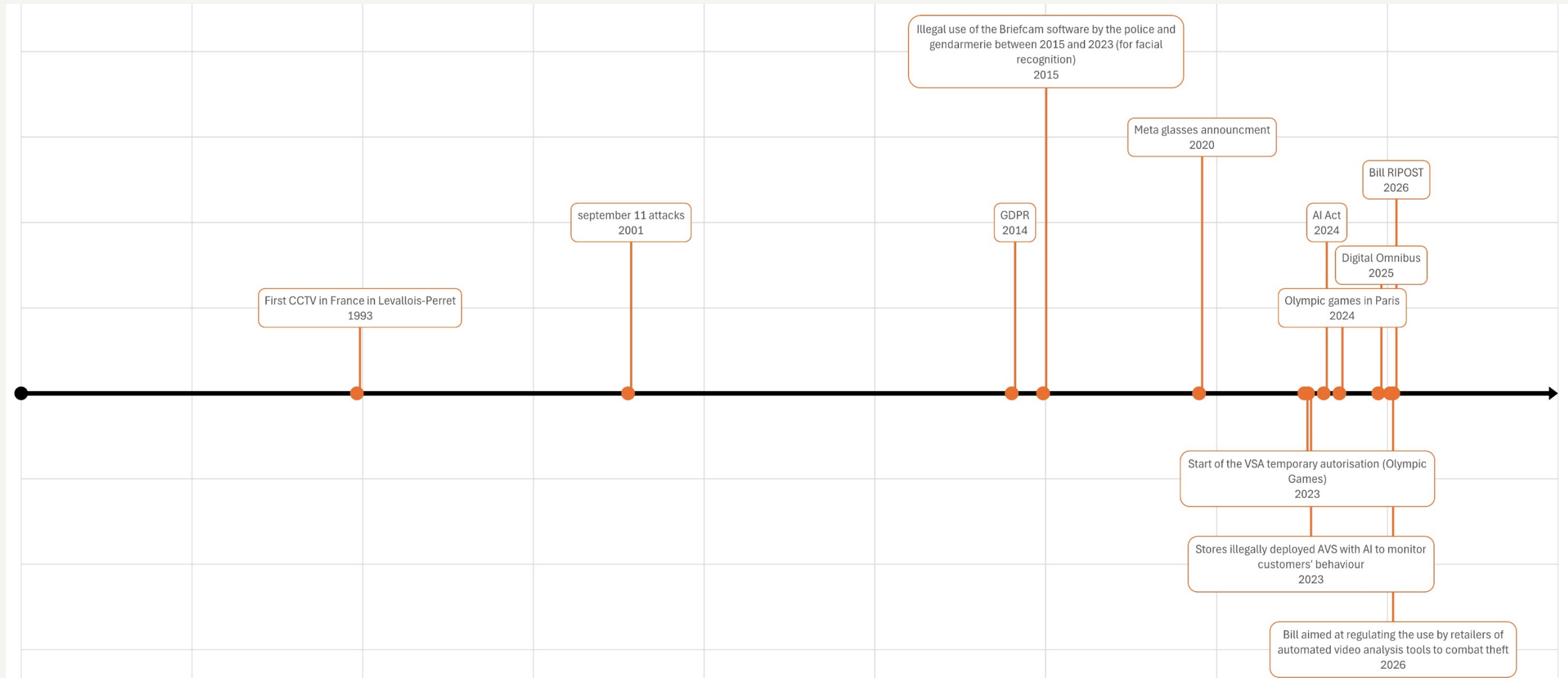
- What is an augmented camera ?
- History
- Case studies
 - 2024 Olympic Games
 - AVS in stores
- Results
- Reglementation vs. reality
- Surveillance techniques' evaluation
- Conclusion and openings

What is an augmented camera ?

- Augmented cameras are video devices (AVS Algorithmic Video Surveillance) that incorporate algorithmic image analysis capabilities (often using AI).
- Can be done in real-time or via recordings
- 3 contexts : public spaces, private spaces and « personal »
- In France, regulated by only one text : the Olympic and paralympic games Act
- Only one EU regulatory text regulates it : the AI Act

[40], [41]

History



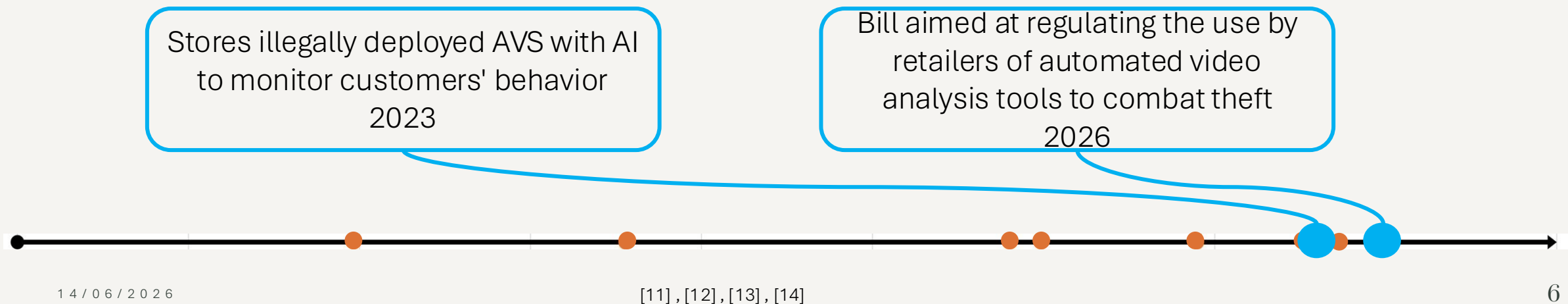
Case study : AVS in the Paris Olympic Games

- Olympic and paralympic games Act (**temporary** experimentation Text)
 - **High risk** AI in the AI Act
 - **Forbidden** to use facial recognition (but clothes recognition as well)
 - **Reconducted** until Olympics 2030
 - RIPOST bill using the **experimentation** in order to entrench the experimentation in the regulation
-
- Regulation **before** the usage of the technology



Case study : AVS in the stores

- **Which data** to train those AI, to demonstrate?
- **Not** in the Olympic and paralympic games Act
- Regulation **after** the first technology's usage
- **Compliant** with AI Act? And the GDPR?



Which results ?

Public CCTV in France: from the French Court of Auditors (Cour des comptes)

- Used to support investigations and public-order operations.
- **No overall correlation** between the presence of CCTV cameras (the level of street crime, or crime resolution rates)

➔ Highlighted the **lack of rigorous evaluation methods** and called for more evidence-based assessments of CCTV effectiveness.

Algorithmic Video Surveillance (AVS) Olympic Games 2024

- The experiment produced **mixed results**
- **Good to detect** : intrusion in restricted areas, wrong-way vehicle detection and to monitor controlled environments (metro corridors, etc.)
- **Significant limitations** : high FPR, difficulties analysing dense crowds and to detect abandoned objects in complex environments

➔ Concluded that the limited duration of the experiment and the small number of real incidents **prevented definitive conclusions** regarding overall effectiveness.

[16] , [17] , [18], [30], [31], [32]

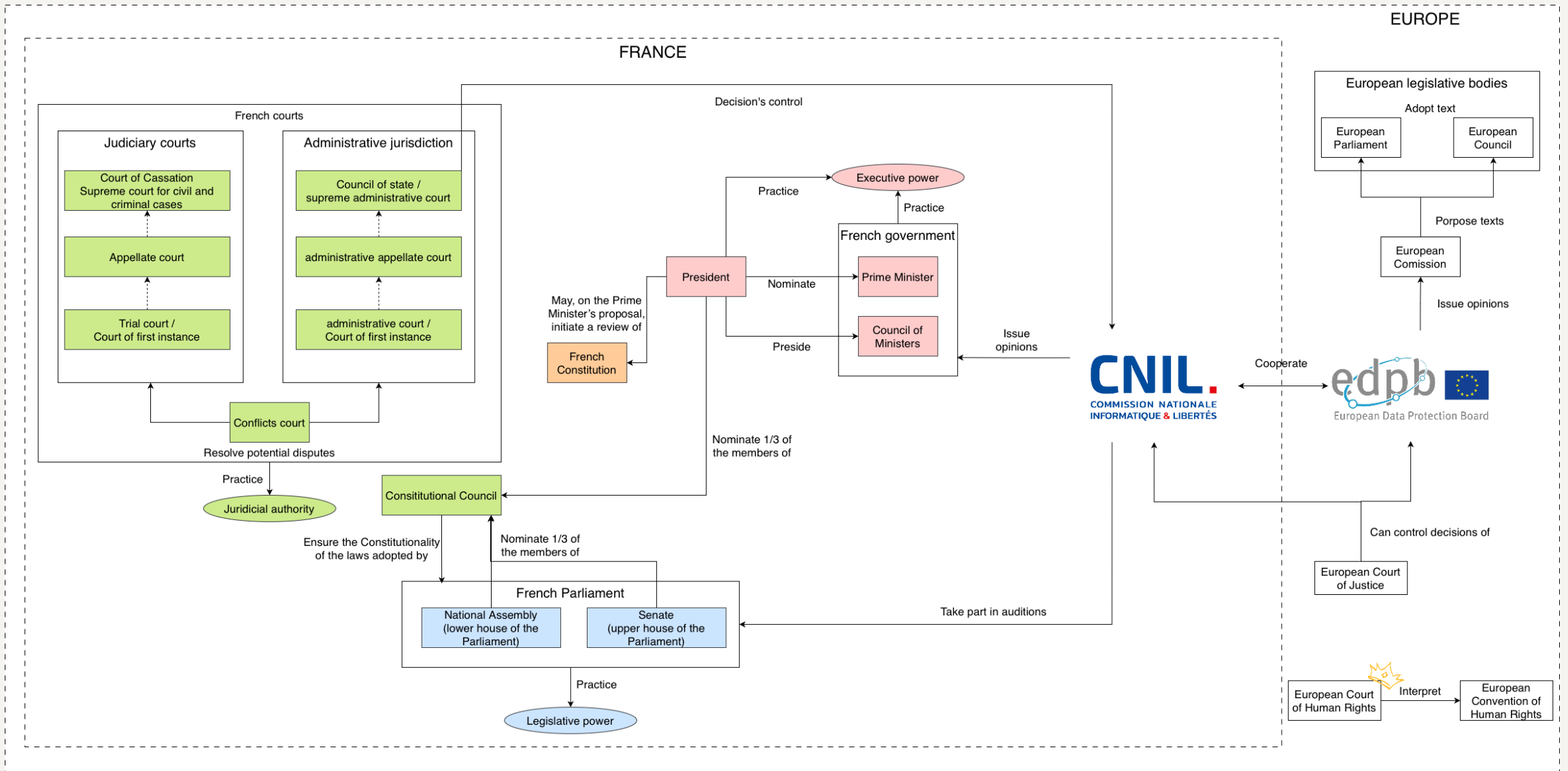
Regulation vs. Reality

- **Multiple texts** (including bills) to regulate the practices
- The **temporary** might be set in stone at some point (Olympic Games exception)
- **Context** dependent

- However, most of the AVS are **illegal**
- **Technology driven** → some illegal usages can be done (what about « personal » AVS such as Meta glasses ?)

Regulation vs. Reality

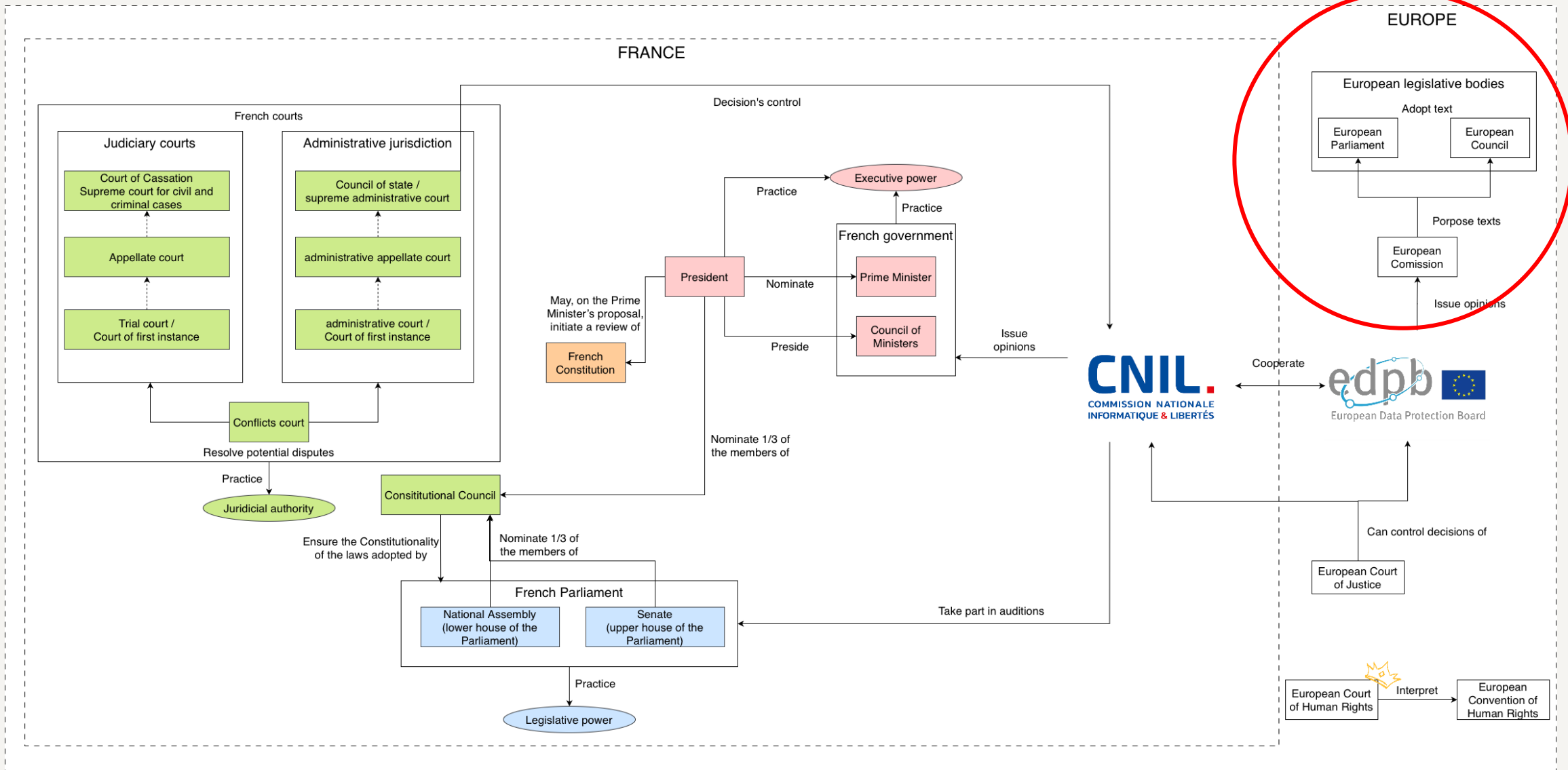
Regulatory context



Regulation vs. Reality

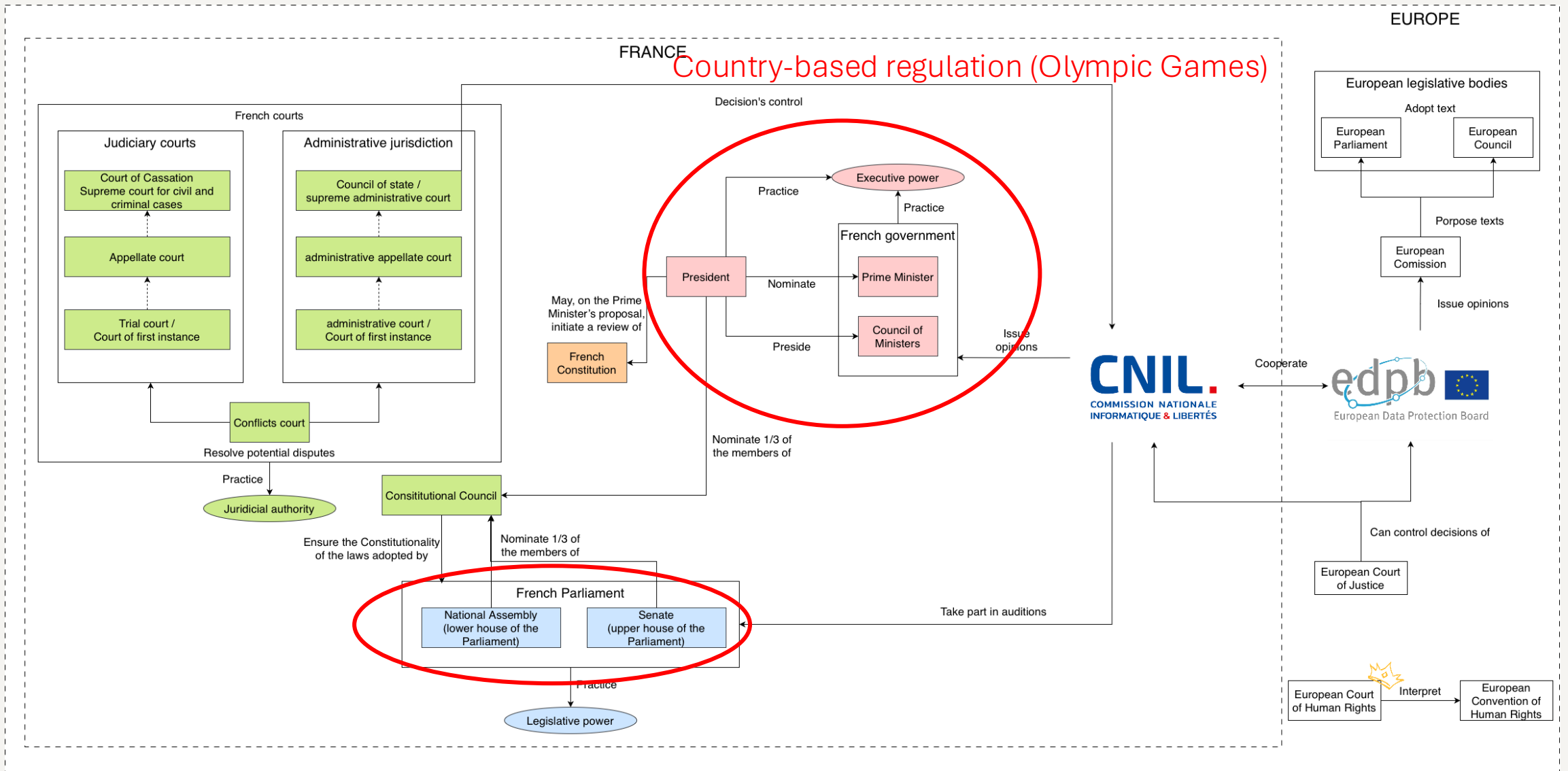
Who propose regulation ?

« Upstream » regulation (AI Act)



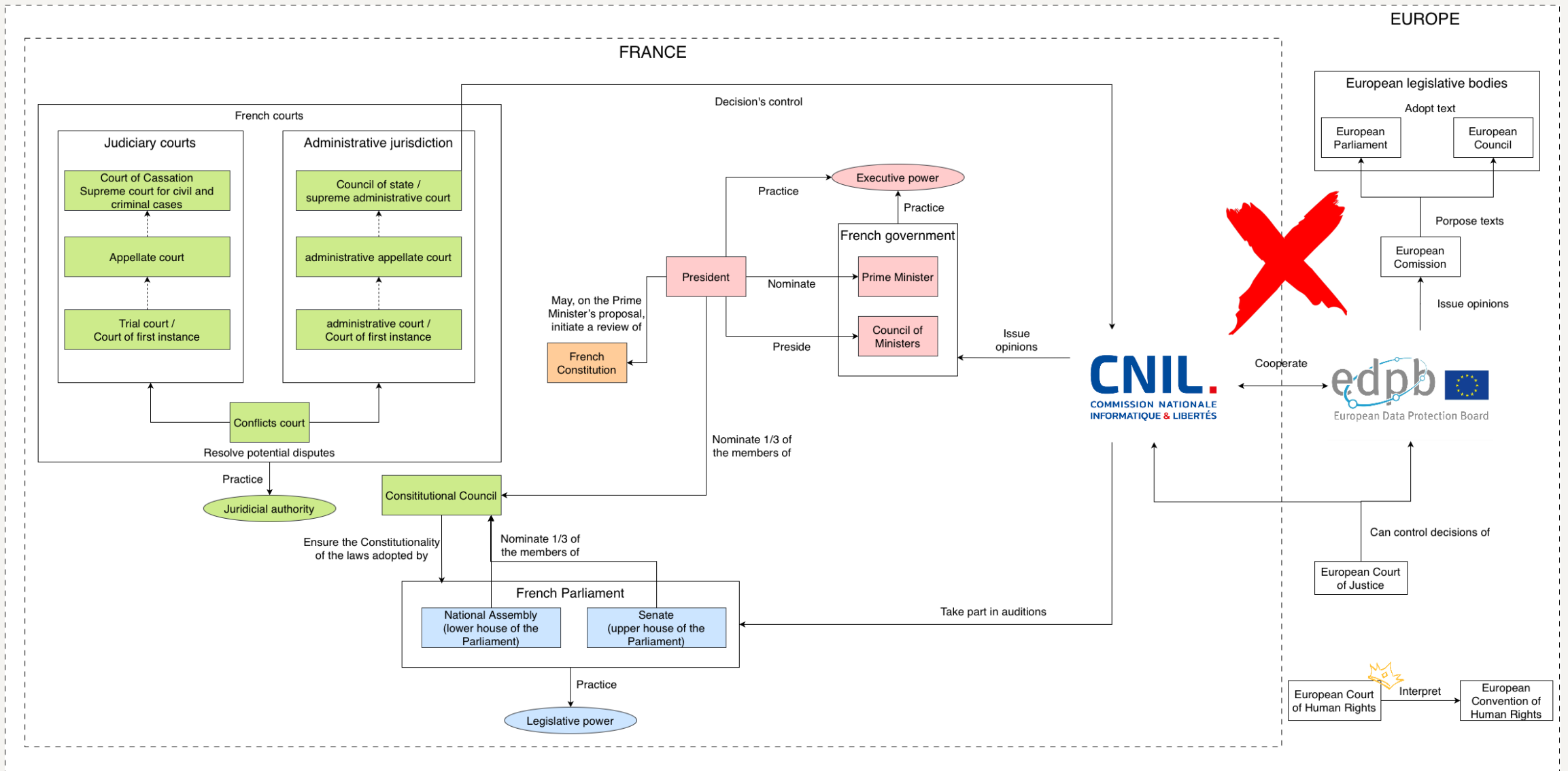
Regulation vs. Reality

Who propose regulation ?



Regulation vs. Reality

Who propose regulation ?



Surveillance techniques' evaluation

- Legality (text regulating the technique)
- Independant supervision
- Transparency
- Remedies
- Impacted fundamental right(s)
- Success rate (technological)
- Any report (other domain)
- Possibility of connection with other database (ease)
- Use of AI (ease)

Credit : Some of those criterias come from Elia Verdon's work (big thanks to her)

C. Collect or G. Generation	1. Target (who)
	2. Duration (when)
	3. Materiality (how)
	4. Type (where and what)
S. Storage	1. Duration (when)
	2. Acces (who)
	3. Storage (where)
	4. Materiality (how)
A. Analyse	1. Date (when)
	2. Treatment (where)
	3. Treatment (who)
	4. Goals (what)

Conclusion and openings

- Numerous criterias
- Context dependent
- (At best) mitigated results
- Lack of assessment methods on those technologies
- Different regulatory texts (when available)
- Technology driven world and regulation
- Acceleration this past decade
- AVS is only a small portion of surveillance techniques



Source : "*Madagascar*", Dreamworks

Thank you for your attention !



If you want to download the presentation and the bibliography

https://github.com/Oraclia/SSN_2026/blob/main/README.md



Source : "Madagascar", Dreamworks